INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
SALT LAKE CAMPUS
(Affiliated to UEM Kolkata and Approved by AICTE)

Degree: BCA, Stream: BCA

Term – I, Second Semester Examination, February - 2024

Subject Code: BCAAE201

Subject Name: Discrete Structure

Full Marks:30 Duration: 1 Hour

Part - A
Attempt 5 questions
Each question carries 2 marks (2 × 5)

1. Define simple graph. 2

or

Define regular graph. 2

2. What is a pendant vertex? 2

or

What is an isolated vertex? 2

3. Draw a graph which is simple and regular. 2

or

Draw a regular graph which is not simple. 2

4. Define Power Set with example. 2

or

What is an infinite set. Give example of 2 different type of infinite sets. 2

5. If U={1,2,3,4,5} and A={1,2},find (A')'. 2

or

Find AXB where A={1,2} and B={a,b}. 2

Part - B
Attempt 2 questions
Each question carries 5 marks (5 × 2)

6. Show that: 5
   1) $A - (B \cup C) = (A - B) \cap (A - C)$
   2) $A - (B \cap C) = (A - B) \cup (A - C)$

or

In an examination of 90 students,65 passed in Physics,50 passed in Maths and 35 passed in both Physics and Maths. Calculate how many students passed exactly in one of the two subjects. 5

7. If A={1,3,5},B={1,3,8} find 5
   1) $A \times B$
   2) $B \times A$
   3) $(A \times B) - (B \times A)$
   4) $A \Delta B$

or

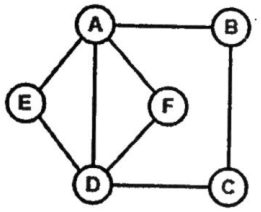
In a class of 50 students,15 read Physics,20 read Chemistry and 20 read Maths.3 read Physics and Chemistry,6 read Chemistry and Maths and 5 read Physics and Maths.7 read none. How many students read all the subjects? 5

Part - C

Attempt 1 question

Each question carries 10 marks (10 × 1)

8. i. Write a short note on Hamiltonian graph. 10
   ii. Check whether the following graph is Hamiltonian or not:



or

i. Write a short note on Euler graph. 10
ii. Check whether the following graph is Euler or not:

